Beyond Carbon Accounting: Why AI Sustainability Governance Must Begin at the Data Layer

I. Introduction

In 2023, Google's greenhouse gas emissions rose 13% year-over-year. Microsoft's increased by over 29% between 2020 and 2024. Both companies attributed the surge primarily to artificial intelligence infrastructure, and both, notably, had standing net-zero commitments at the time. This is not a failure of corporate intention. It is a failure of the frameworks designed to govern it.

The environmental footprint of generative AI operating within cloud infrastructure has become one of the defining sustainability challenges of the current decade, and it is one that existing regulatory and accounting mechanisms are structurally unequipped to address. Training a single large foundation model (estimates suggest GPT-4's training run consumed between 51 and 62 million kilowatt-hours of electricity and may have emitted upward of 15,000 metric tons of CO_2 equivalent) represents an environmental event of meaningful scale. Yet that training run, hosted on cloud infrastructure, falls into Scope 3 for most enterprises under the GHG Protocol, effectively rendering it invisible in corporate sustainability reporting. And training, counterintuitively, is not the most consequential phase. Inference at scale (serving 700 million weekly users, processing an estimated one billion queries per day) generates cumulative environmental impacts that dwarf a single training run over any sustained deployment period, and remains even more systematically underreported.

The standard policy response to this problem has been to push for better carbon disclosure: more granular Scope 3 reporting, mandatory model cards, extended producer responsibility for hardware. These interventions are necessary but not sufficient. They address the symptom (inadequate accounting of emissions after the fact) without engaging the structural cause: the absence of governance at the point where environmental cost is actually determined. That point is not the data center. It is the dataset.

This paper argues that the computational lifecycle of generative AI systems contains a largely ungoverned upstream layer (the data acquisition, curation, and preprocessing phase) that functions as the primary lever for environmental cost without being recognized as such by any current regulatory framework. Decisions made at the data layer, including the size, composition, deduplication strategy, and provenance of training datasets, have direct and significant implications for the energy intensity of model training, yet are made entirely outside the scope of environmental governance. Closing the regulatory gap that allows AI systems to externalize their environmental costs therefore requires more than expanded carbon accounting. It requires reconceptualizing data governance as a form of sustainability governance, integrating environmental impact assessment into the earliest stages of the AI development pipeline rather than attempting to measure and offset damage after it has occurred.

The paper proceeds in four parts. Section II maps the full computational lifecycle of generative AI systems hosted on cloud infrastructure and identifies the specific failure modes of existing environmental frameworks at each stage. Section III establishes generative AI (and large foundation models in particular) as the critical case that exposes these failure modes most acutely, examining the disclosure gap across major hyperscalers. Section IV develops the core argument: that data governance interventions function as de facto sustainability interventions, and proposes three mechanisms through which a data-layer governance framework could be operationalized. Section V outlines the principles of a purpose-built AI sustainability framework, addresses the jurisdictional coordination challenge, and situates the proposal relative to existing regulatory attempts in the EU, US, and beyond.

II. The Computational Lifecycle: What Current Frameworks Miss

Any credible governance framework must first define the system it governs. For the purposes of this analysis, the computational lifecycle of a generative AI system hosted on cloud infrastructure spans five interdependent phases: data acquisition and curation; model training; fine-tuning and alignment; inference at scale; and hardware manufacturing and end-of-life. Each phase generates distinct and material environmental costs. Each is inadequately governed by existing frameworks. And critically, the failure of current mechanisms at each phase is not coincidental. It is a structural consequence of frameworks designed for a different industrial paradigm.

The Data Layer

The environmental costs of the data layer are the least visible and the least studied. Web-scale data acquisition for foundation model training involves the continuous crawling, downloading, and storage of hundreds of billions of documents. The compute required for dataset preprocessing (deduplication, quality filtering, tokenization, and formatting) is itself non-trivial, routinely running on GPU clusters for days or weeks prior to the training run proper. The storage infrastructure required to maintain these datasets (often multiple petabytes per model family) operates in hyperscale data centers that consumed an estimated 460 terawatt-hours of electricity globally in 2024, generating approximately 182 million metric tons of CO_2 equivalent (IEA, 2024). No current regulatory instrument requires the attribution of any portion of this footprint to specific models or training pipelines. The data layer is, in governance terms, a void.

The Training Layer

Training is the phase most frequently cited in the literature and most commonly addressed in voluntary disclosure. Even here, however, the picture is incomplete. The energy intensity of pre-training a large foundation model is substantial: estimates place the electricity consumption of GPT-4's training run between 51 and 62 million kilowatt-hours, with associated emissions heavily dependent on the carbon intensity of the grid region in which training occurred, a factor that can vary by as much as 13-fold between Azure cloud regions. This regional dependence is not a minor methodological consideration; it means that a model trained in a fossil-fuel-intensive region may carry carbon costs an order of magnitude higher than the same model trained under renewable energy, yet neither outcome is captured with consistency in existing disclosure frameworks.

Under the GHG Protocol, cloud-hosted training falls into Scope 3 for the enterprise purchasing compute, specifically Category 1, Purchased Goods and Services. This is not an incidental classification; it is a structural feature of a framework originally designed around physical supply chains, in which the responsible emitter was the organization that owned or controlled the source of emissions. As a landmark analysis published in *Nature Climate Change* observed, the migration of IT workloads to the cloud systematically shifts emissions that would previously have been captured under Scope 1 or Scope 2 into Scope 3, where reporting is voluntary in most jurisdictions and methodologically inconsistent across firms. The practical consequence is that an enterprise can route its most emissions-intensive AI workloads through a cloud provider and report significantly reduced emissions, not because emissions have been reduced, but because accountability has been redistributed to a category where it remains largely invisible.

The Inference Layer

Inference (the continuous, real-time serving of model outputs to end users) is the phase least addressed in current policy discussions and arguably the most consequential over time. Training a model is a one-time event; inference is perpetual. Scaling a deployed foundation model to hundreds of millions of users generates cumulative emissions that dwarf the training run across any meaningful deployment horizon. Research suggests that inference already accounts for over 80% of total AI electricity consumption in production deployments. A single H100 GPU draws approximately 700 watts under full compute load; an eight-card server node consumes between 10 and 12 kilowatts; a training and inference cluster of 10,000 GPUs requires 10 to 15 megawatts of continuous power, sufficient to supply a small city. AI workloads are now estimated to account for 15 to 20% of total global data center electricity demand, with projections suggesting this share could approach 47% by end of 2025 at current growth trajectories (IEA, 2024). Despite this, no major environmental reporting standard specifically addresses inference-phase emissions as a separate, mandatory disclosure category.

The Hardware Layer

The embodied carbon of AI accelerator hardware constitutes a material and systematically underreported component of the lifecycle footprint. NVIDIA's own Product Carbon Footprint assessment, published in 2025, places the embodied emissions of a single HGX H100 baseboard (containing eight H100 SXM cards) at 1,312 kg CO_2 equivalent, with semiconductor components (including the main die and high-bandwidth memory) accounting for approximately 67% of that total. These emissions arise from the fabrication of chips at 5 to 7 nanometer process nodes at concentrated foundries (primarily TSMC in Taiwan), processes that are extraordinarily energy- and water-intensive, and that are geographically distant from both the organizations reporting emissions and the regulatory jurisdictions in which those organizations operate. A Cornell University study projects that at current growth trajectories, AI systems could emit 24 to 44 million metric tons of CO_2 annually by 2030. Embodied hardware emissions, routinely excluded from model-level carbon assessments, represent a portion of that figure that current governance frameworks are not structured to capture.

The Hyperscaler Intermediary Problem

Threading through all four phases is a structural feature of the cloud AI ecosystem that amplifies each of the above failure modes: the intermediary role of hyperscalers. AWS, Microsoft Azure, and Google Cloud collectively host the overwhelming majority of AI training and inference workloads. Their position in the value chain means that the emissions generated by AI workloads are, for most enterprises, categorically invisible. They are Scope 3 by definition, reported (if at all) in the hyperscaler's own sustainability disclosures rather than the enterprise's. The quality of those disclosures varies substantially. As of mid-2025, AWS's Customer Carbon Footprint Tool still does not include Scope 3 emissions, a gap that allows the tool to return near-zero results for compute-intensive workloads while remaining nominally compliant with current reporting requirements. Google provides both market-based and location-based figures, offering more granular accountability. Azure offers market-based results as the default, with location-based figures available but not prominently surfaced. This inconsistency is not merely an inconvenience for sustainability analysts; it is a governance failure that makes system-level accountability for AI emissions structurally impossible under the current framework.

Taken together, these failure modes describe a coherent pattern: a lifecycle that generates significant and growing environmental costs, governed by frameworks that are fragmented by scope, inconsistent by jurisdiction, and systematically blind to the phase (the data layer) where those costs are first determined.

III. Generative AI as the Critical Case

The failure modes described in Section II are not unique to generative AI. Any large-scale cloud workload (genomics pipelines, financial modeling, real-time logistics) exposes similar gaps in the current governance architecture. What distinguishes generative AI, and foundation models in particular, is the degree to which these failure modes are concentrated, compounded, and accelerating. Generative AI does not merely fall through the cracks of existing frameworks; it structurally maximizes the exposure created by each one.

The Scale Problem

The compute requirements of frontier foundation model training represent a categorically different order of magnitude from prior machine learning paradigms. Where earlier deep learning systems could be trained on a single GPU cluster over days, training a frontier foundation model now involves tens of thousands of accelerators running for months, with energy consumption scaling nonlinearly as models grow in parameter count and training data volume. The transition from GPT-3 to GPT-4 alone represented an estimated 40- to 48-fold increase in electricity consumption. This scaling dynamic has not plateaued. Goldman Sachs projects that data center electricity consumption will more than double between 2022 and 2030, with generative AI as the primary driver. The IEA projects that global data center electricity demand could reach nearly 1,000 terawatt-hours annually by 2030, exceeding Japan's current total national consumption. At current trajectories, AI workloads could generate between 24 and 44 million metric tons of CO_2 annually by 2030 (Cornell University, 2024).

What makes this a governance problem, rather than merely an engineering challenge, is the rate at which these emissions are being generated relative to the rate at which accountability frameworks are being developed. The gap is not closing. It is widening.

The Rapid Iteration Problem

Foundation model development does not produce a single artifact. It produces a continuous succession of pre-trained base models, fine-tuned variants, alignment iterations, and deployment-specific adaptations. Each iteration carries its own training energy cost. Each version of a model family (the base model, its instructiontuned variant, its reinforcement-learning-from-human-feedback iteration) represents a discrete training run that multiplies the lifecycle footprint of the system as a whole. Because current disclosure frameworks focus on model-level reporting where they report at all, the cumulative emissions from a model family's full development pipeline are systematically understated. A foundation model's disclosed carbon footprint, where one exists, typically reflects only the final training run, not the experimental runs, ablation studies, or intermediate checkpoints that preceded it.

The Disclosure Gap: A Comparative Analysis

The three major hyperscalers present a revealing case study in the inconsistency that results from voluntary, non-standardized disclosure. Google's 2023 Environmental Report disclosed that total GHG emissions increased 13% year-over-year, attributing the rise primarily to data center energy consumption driven by AI infrastructure expansion. Google's emissions had risen approximately 48% from its 2019 baseline, a reversal so significant that the company formally stepped back from prior claims of carbon neutrality. Microsoft's 2024 Sustainability Report disclosed that emissions had grown 29.1% above its 2020 baseline, driven by a 30.9% increase in Scope 3 emissions from data center expansion. Its energy use has increased 168% since 2020. Amazon reported a 3% emissions reduction in its most recent available data, though the company simultaneously flagged AI infrastructure expansion as the primary expected source of future emissions growth, and Amazon's own Scope 3 data remains absent from its Customer Carbon Footprint Tool as of mid-2025.

These three disclosures are not comparable. They use different baseline years, different methodological treatments of renewable energy certificates, and different coverage of Scope 3 categories. A 2025 analysis by NewClimate Institute and Carbon Market Watch, reviewing all five major tech companies under the Corporate Climate Responsibility Monitor, concluded that their GHG emissions targets had effectively "lost their meaning and relevance." The analysis further noted that at least one-third of tech sector emissions footprints derive from computer hardware manufacturing, a category that none of the five companies has set specific reduction targets for.

This inconsistency is not a temporary gap awaiting resolution through voluntary improvement. It is the predictable outcome of a governance architecture that allows each actor to define its own reporting boundaries, methodology, and baseline, with no standardized treatment of the emissions categories most material to AI systems.

The Green AI Paradox

The most acute expression of the regulatory failure in generative AI is what might be termed the green AI paradox: the simultaneous deployment of foundation models as tools for climate modeling, grid optimization, and emissions monitoring, and as sources of the very emissions those models are being used to address. The same hyperscalers making net-zero commitments are also the primary infrastructure providers for frontier model training at scale, with the emissions from that training routinely managed through offset purchases rather than source reduction. Google has committed to purchasing carbon removal credits from multiple vendors, including contracts with direct air capture and enhanced weathering startups. Microsoft has purchased more than 30 million tonnes of high-quality carbon removal, representing approximately 80% of the global market as of late 2025. These are not negligible investments. But they are a response to rising emissions, not a substitute for governance that prevents those emissions from arising in the first place.

The paradox extends to the data layer. The datasets used to train foundation models are assembled, in significant part, from the accumulated knowledge of environmental science, climate research, and sustainability literature. The curation of those datasets (the decisions about what to include, at what scale, with what preprocessing) carries energy costs that are completely ungoverned. No framework currently requires a model developer to assess or disclose the environmental cost of data acquisition and curation as a precondition for training. The result is a system in which AI is simultaneously positioned as a sustainability solution and exempted from the sustainability governance applied to the industries it is meant to help decarbonize.

This paradox is not a communications failure. It is a structural consequence of treating the data layer as outside the scope of environmental governance, which is precisely the gap that the following section proposes to close.

IV. Data Governance as a Sustainability Intervention

The analysis in Sections II and III establishes that the primary failure of current AI sustainability governance is not a measurement failure, though measurement is inadequate. It is a scoping failure: a structural decision, embedded in the architecture of existing frameworks, to treat the data layer as outside the domain of environmental governance. This section argues that this scoping decision is not merely incomplete; it is incoherent. Data governance decisions are, by their nature, environmental decisions. The question is not whether they should be recognized as such, but how governance frameworks can be designed to operationalize that recognition.

The argument rests on a straightforward empirical observation: the size, composition, and curation methodology of a training dataset are among the most significant determinants of a model's energy cost, preceding and constraining every downstream phase of the lifecycle. A model trained on a larger dataset requires more compute. A dataset with higher noise levels requires more training iterations to converge. A dataset with poor deduplication produces models that learn redundant representations, wasting compute on signal the model has already internalized. These are not marginal efficiency considerations. They are primary levers, and they are currently pulled by dataset curators, data engineers, and ML researchers operating entirely outside any environmental governance framework. Three mechanisms translate this observation into a governance architecture.

Mechanism 1: Dataset Efficiency as Environmental Policy

The data-centric AI paradigm, advanced most prominently through the work of Andrew Ng and subsequent academic literature, represents a methodological reorientation away from model-centric scaling (the practice of achieving performance gains by increasing model size and training data volume) toward systematic improvement of dataset quality as the primary lever for model improvement. Research within this paradigm has demonstrated consistently that carefully curated, higher-quality datasets can match or outperform larger, noisier alternatives on downstream benchmarks, while requiring significantly less compute to train upon. A comprehensive analysis of production ML systems suggests that approximately 80% of performance variance is attributable to data quality factors, with model architecture and optimization accounting for the remaining 20%, a distribution that directly inverts the priorities of most current AI development practice and, by extension, most current AI environmental governance discussion.

From a sustainability governance standpoint, the data-centric AI literature provides the evidentiary foundation for a regulatory claim that would otherwise be difficult to assert: mandating higher data quality standards is not merely a fairness or accuracy intervention. It is an emissions intervention. Regulatory requirements compelling dataset deduplication, noise filtering, and quality validation at scale would, if implemented, reduce the compute required to achieve equivalent model performance, thereby reducing the energy intensity of training without requiring any change to model architecture, hardware, or grid energy mix. This is not a theoretical benefit. It is a direct, quantifiable reduction pathway that current frameworks have no mechanism to pursue because they govern the training run, not the data that precedes it.

Mechanism 2: A Tiered Licensing Framework for Training Data

The second mechanism addresses a deeper structural problem: the absence of any accountability system that can attribute environmental cost to specific data inputs. The current state of training data governance is characterized by opacity that would be considered extraordinary in any other regulated industry. Foundation model developers routinely decline to disclose the composition, sources, or scale of their training datasets, citing competitive sensitivity. Where disclosures exist (typically in technical reports accompanying model releases) they are voluntary, inconsistent in methodology, and contain no environmental cost attribution whatsoever. The data layer is, in accountability terms, a black box.

A tiered licensing framework for AI training data, modeled on the rights management architecture of the music industry, offers a governance mechanism that can introduce accountability at the data layer without requiring blanket disclosure of proprietary dataset composition. Under this framework (developed as part of an original AI sustainability governance architecture) training data is classified into three tiers based on source type, scale of use, and associated environmental cost attribution requirements.

The first tier covers publicly licensed and open-access data: materials released under Creative Commons licenses, government open data, and other sources whose terms of use explicitly permit AI training. For Tier 1 data, environmental cost attribution is achieved through mandatory disclosure of dataset volume and preprocessing methodology in a standardized machine-readable format, analogous to the Data Cards framework developed by Google Research and presented at the ACM FAccT conference. Disclosure at this tier is relatively low-burden and establishes the baseline accounting record for the training pipeline.

The second tier covers commercially licensed data acquired through direct agreements with content creators, publishers, or data aggregators. This tier draws the most direct analogy to music industry rights management: just as a streaming platform must report performance data to rights holders to calculate royalty obligations, a model developer acquiring Tier 2 training data would be required to disclose the volume, preprocessing cost, and proportional use of that data in the training pipeline to a designated environmental accounting authority. Licensing agreements for Tier 2 data would include standardized environmental cost disclosure clauses as a condition of use, creating a contractual mechanism for attribution that does not depend on voluntary compliance.

The third tier covers data acquired through large-scale web crawling (the dominant method for assembling the datasets used to train most frontier foundation models) where attribution to original sources is technically infeasible at scale. Tier 3 data carries the highest disclosure requirements precisely because it represents the largest share of training data volume and the greatest accountability gap. Developers using Tier 3 data would be required to disclose crawler energy consumption, dataset deduplication rates, filtering methodology, and total token count in a mandatory pre-training environmental impact assessment, filed prior to the commencement of training and subject to audit by a designated regulatory body.

The music industry analogy is instructive not only as a structural model but as a proof of concept. Rights management in music operates across an extraordinarily complex ecosystem of creators, publishers, distributors, and platforms, with attribution chains that span decades and jurisdictions, and it functions (imperfectly but operationally) through a combination of licensing agreements, collective rights organizations, and mandatory reporting. AI training data governance faces a similarly complex attribution problem. The institutional infrastructure for managing it at scale already exists in adjacent domains. The governance decision is whether to build toward it.

Mechanism 3: Compute-Aware Data Standards and Pre-Training Environmental Assessment

The third mechanism addresses the timing problem inherent in current disclosure frameworks: they require reporting after emissions have occurred, when no governance intervention can reduce the environmental cost of a training run that has already happened. A compute-aware data standard would shift the point of regulatory engagement upstream, requiring environmental impact assessment at the dataset curation and model architecture stage (before the training run begins) rather than attempting to account for and offset damage after the fact.

Under this standard, any training run exceeding a defined compute threshold (expressed in floating-point operations, GPU-hours, or equivalent energy units, calibrated to cover frontier model training while exempting research and fine-tuning workloads) would require a pre-training environmental impact assessment. The assessment would document dataset composition, total token count, deduplication and filtering methodology, estimated training energy under a defined grid carbon intensity scenario, and planned inference deployment scale. This assessment would be submitted to a regulatory body prior to training commencement, creating an accountable record and enabling regulatory review of high-impact training runs before they proceed.

This is not without precedent in adjacent domains. Environmental impact assessment requirements for large industrial projects (codified in the US under the National Environmental Policy Act and in the EU under the Environmental Impact Assessment Directive) already establish the principle that significant environmental interventions require prospective review rather than retrospective accounting. The argument for extending this principle to frontier AI training is straightforward: a training run consuming tens of millions of kilowatt-hours of electricity is a significant environmental event by any reasonable definition, and the fact that it occurs in a data center rather than a construction site does not exempt it from the logic that motivated impact assessment requirements in the first place.

The three mechanisms are designed to be mutually reinforcing. The tiered licensing framework creates the institutional infrastructure for data-layer accountability. The dataset efficiency standard provides the technical basis for translating data governance requirements into quantifiable emissions reductions. The pre-training environmental assessment creates the regulatory touchpoint that makes both the licensing framework and the efficiency standard enforceable. None of the three mechanisms is sufficient alone. Together, they constitute a governance architecture that treats the data layer as what it is: the first and most consequential node in the AI sustainability chain.

V. Toward a Purpose-Built Regulatory Framework

The governance mechanisms proposed in Section IV do not function in a vacuum. Their effectiveness depends on an institutional architecture capable of implementing them consistently across jurisdictions, enforcing them against actors with strong competitive incentives to resist disclosure, and calibrating them in ways that do not inadvertently concentrate AI development among the small number of organizations capable of bearing compliance costs. This section develops the design principles for such a framework, proposes hyperscalers as the appropriate regulated intermediary, addresses the most significant structural tradeoff the framework entails, and situates the proposal against the existing regulatory landscape.

Design Principles

A purpose-built AI sustainability framework must be built around four foundational principles that distinguish it from the incremental adaptations of existing frameworks that have characterized governance attempts to date.

The first principle is *lifecycle completeness*. Any framework that scopes out the data layer, the inference phase, or the hardware manufacturing chain (as all current frameworks do, to varying degrees) will produce an incomplete and gameable accounting of AI's environmental footprint. Lifecycle completeness does not require identical disclosure intensity at every phase; it requires that no phase be exempt from the framework's coverage by design. The regulatory burden at each phase should be proportional to both the materiality of the environmental cost and the feasibility of measurement, but the framework's boundary must encompass the full lifecycle.

The second principle is *jurisdictional coordination*. Foundation model training is inherently transnational. A model may be developed by a US company, trained on compute infrastructure distributed across data centers in multiple countries, using datasets assembled from sources in dozens of jurisdictions, then deployed globally through cloud infrastructure subject to yet another set of national regulatory regimes. Unilateral regulation by any single jurisdiction cannot achieve consistent accountability for a system this geographically distributed. This does not make regulation impossible; it makes multilateral coordination a precondition for regulatory effectiveness. The model for such coordination exists in international aviation emissions governance through CORSIA, developed by the International Civil Aviation Organization and now covering approximately 99% of international aviation CO_2 emissions across 128 participating states as of 2025. CORSIA is instructive not as a model to be replicated (aviation emissions are operationally simpler to attribute than AI training emissions) but as a proof of concept that a globally distributed, commercially competitive industry can be brought under a functionally unified emissions accounting standard through multilateral institutional design. An equivalent body for AI environmental governance (potentially housed within an existing institution such as the International Telecommunication Union or established as a dedicated multilateral organization) would be charged with setting the technical standards for compute-threshold triggers, emissions attribution methodology, and disclosure format that individual jurisdictions implement through their own regulatory machinery.

The third principle is *compute-proportional disclosure*. Regulatory burden must scale with the environmental materiality of the activity being regulated. A framework that imposes identical disclosure requirements on a frontier foundation model training run and a fine-tuned task-specific model would be both technically unjustifiable and politically unsustainable. Compute-proportional disclosure (calibrated to training FLOPs, dataset token count, and deployment scale) creates a tiered compliance structure analogous to tiered disclosure requirements in financial regulation, where reporting obligations scale with the size and systemic importance of the regulated entity. This principle also enables the regulatory threshold to remain calibrated to technological development: as compute costs fall and frontier training scales expand, the threshold can be adjusted to maintain consistent coverage of the highest-impact training runs without over-burdening the broader research and development ecosystem.

The fourth principle is *data-layer integration*, which is the core contribution of this paper's framework relative to existing governance proposals. Environmental impact assessment must be triggered at the dataset curation stage, before the training run begins, rather than at the deployment stage, after the environmental cost has been incurred. This principle operationalizes the central argument of Section IV: effective AI sustainability governance requires governing the upstream decisions that determine environmental cost, not only the downstream emissions that result from them.

The Hyperscaler Intermediary

The most significant institutional design challenge for any AI sustainability framework is enforcement: who is the accountable actor, and through what mechanism can accountability be made operational at scale? The three-tier licensing framework and pre-training assessment mechanism proposed in Section IV identify the model developer as the primary accountable party. This is correct in principle but faces a practical obstacle: model developers range from small research organizations to large corporations, operate across multiple jurisdictions simultaneously, and have strong competitive incentives to minimize the scope and granularity of environmental disclosures.

Hyperscalers offer a solution to this enforcement problem. AWS, Microsoft Azure, and Google Cloud collectively host the overwhelming majority of frontier AI training workloads. Their position as infrastructure providers for the AI ecosystem makes them natural regulatory intermediaries, analogous to the role financial institutions play in anti-money laundering compliance: not the primary actors subject to the underlying regulatory obligation, but the chokepoint through which regulated activity flows, and therefore the practical locus of enforcement. Under this model, hyperscalers would be required, as a condition of providing compute infrastructure for training runs exceeding the defined threshold, to collect and report the pre-training environmental assessment data described in Section IV, verify the compute usage metrics underlying emissions calculations, and transmit standardized emissions records to a designated regulatory body. This model draws directly on the precedent established by the EU's General Data Protection Regulation, which designates cloud providers as data processors subject to specific compliance obligations when handling personal data on behalf of data controllers. The same institutional logic applies to environmental governance: the provider of the infrastructure through which regulated activity occurs bears a defined set of compliance obligations as a condition of market participation.

Addressing the Concentration Risk

The most substantive objection to the framework proposed here is that stricter data governance and pre-training assessment requirements will raise barriers to entry in AI development, further concentrating model development among well-resourced incumbents, which are the exact organizations whose current dominance has produced the disclosure gap the framework seeks to close. This objection deserves direct engagement rather than dismissal.

The concern is empirically grounded. California's SB 1047, which proposed compute-threshold-based pre-training safety requirements for frontier models, was ultimately vetoed by Governor Newsom in September 2024 in part because of concerns that its compliance burden would disadvantage smaller developers relative to incumbents who could more easily absorb the costs of safety protocols and audits. California's subsequently enacted SB 53, signed in September 2025, applies only to developers with more than $500 million in annual revenue, a threshold that explicitly acknowledges the concentration risk and attempts to scope the regulatory burden to actors with the capacity to bear it.

The framework proposed here incorporates the same logic through its compute-proportional disclosure principle: the highest-burden requirements, including the full pre-training environmental assessment and the Tier 3 data disclosure obligations, apply only to training runs exceeding defined thresholds that correspond to frontier model development. Research-scale training, fine-tuning, and the vast majority of task-specific model development would fall outside the framework's highest-burden tier. More fundamentally, the data-layer governance mechanisms proposed here (particularly the dataset efficiency standards in Mechanism 1 of Section IV) create incentive structures that actively benefit smaller developers. If higher data quality is demonstrated to reduce compute requirements for equivalent model performance, the developers most likely to benefit are precisely those who cannot afford to scale their way to capability through brute-force data accumulation. Dataset efficiency requirements would, in this sense, be more burdensome for incumbents who have built their development pipelines around large-scale web crawling than for smaller developers pursuing data-centric approaches.

Where Existing Frameworks Fall Short

The proposed framework must be situated against the existing regulatory landscape to make the "purpose-built" claim legible. Three regulatory instruments are most directly relevant.

The EU AI Act (Regulation 2024/1689), which entered into force in August 2024, represents the most ambitious attempt to date to incorporate environmental considerations into AI governance. The European Parliament's position, published in June 2023, contained substantive provisions: mandatory energy and resource consumption reporting for foundation models, sustainability impact assessments, and an explicit principle requiring AI to be developed in a "sustainable and environmentally friendly manner." These provisions were substantially diluted during trilogue negotiations. The final text requires providers of general-purpose AI models to disclose "known or estimated energy consumption," a formulation that applies exclusively to the GPAI category, omits water and hardware lifecycle impacts entirely, contains no data-layer requirements, and focuses on training while leaving inference-phase emissions unaddressed. Article 40 establishes a process for developing harmonized standards for resource efficiency, but the first Commission report on progress is not due until August 2028. The EU AI Act is not a failure of ambition at the legislative drafting stage; it is a failure of follow-through at the negotiation stage, where environmental provisions were traded against innovation concerns, producing exactly the voluntary-and-incomplete architecture that characterizes the pre-existing landscape.

California's SB 1047, ultimately vetoed, is instructive for a different reason. Its compute-threshold mechanism (which would have triggered pre-training safety obligations for models meeting defined FLOP and cost thresholds) established a governance logic that this framework's pre-training environmental assessment mechanism directly extends. SB 1047's failure was not a rejection of the compute-threshold approach per se. Governor Newsom's veto message explicitly endorsed the principle of pre-training oversight while objecting to the specific liability structure and the absence of federal coordination. The successor legislation, SB 53, applies the same compute-threshold logic but narrows the scope to safety rather than environmental governance, and to large developers rather than the full developer ecosystem. It confirms the political viability of compute-threshold triggers while leaving the environmental dimension entirely unaddressed.

The gap left by both instruments is precisely the gap this framework is designed to fill: a governance architecture that is lifecycle-complete, data-layer-integrated, and jurisdictionally coordinated, not as aspirations for future standards development, but as operational requirements of the framework itself.

VI. Conclusion

The central claim of this paper is straightforward: existing environmental governance frameworks cannot adequately address the sustainability costs of generative AI systems because they were not designed to. The GHG Protocol's Scope 1/2/3 structure was developed for physical supply chains in which the emitting actor owns or controls the emission source. Cloud-hosted AI training systematically redistributes accountability into Scope 3, where it becomes invisible. The EU AI Act's environmental provisions were diluted through negotiation into voluntary codes of conduct and aspirational standards whose first compliance reports are not due until 2028. California's most ambitious AI governance attempt was vetoed. The hyperscalers at the center of the AI infrastructure ecosystem publish sustainability reports with incompatible methodologies, inconsistent Scope 3 coverage, and no data-layer disclosure whatsoever. None of this is accidental. It is the predictable outcome of applying legacy frameworks to a novel industrial system they were never designed to govern.

The two pillars of this paper's argument are designed to address this structural failure at its root. The first pillar (the regulatory gap analysis across the full computational lifecycle) establishes that the failure is not a measurement problem amenable to incremental improvement in existing disclosure frameworks. It is a scoping problem: the data layer, where the environmental cost of AI is first and most consequentially determined, is entirely outside the boundary of every major current governance instrument. The second pillar (the reconceptualization of data governance as sustainability governance) establishes that closing this gap is both technically feasible and institutionally grounded. The data-centric AI literature provides the evidentiary basis for treating dataset efficiency as an emissions reduction pathway. The music industry's rights management architecture provides the institutional precedent for attributing cost across complex, distributed data pipelines. The pre-training environmental impact assessment mechanism has direct analogues in existing environmental law. None of the three governance mechanisms proposed in Section IV requires the invention of new regulatory logic, only the application of existing logic to a domain that has so far escaped its reach.

The stakes of this governance gap are not abstract. Foundation models are rapidly becoming embedded infrastructure: in climate modeling, energy grid optimization, scientific research, and public health surveillance. The environmental governance of AI is therefore not a niche concern within sustainability policy; it is a precondition for the credibility of AI's role as a sustainability tool. A system whose own environmental costs are systematically ungoverned and underreported cannot be trusted as the instrument through which other industries are held to environmental account.

For the consultants, standards bodies, and policymakers who constitute this paper's primary audience, the practical implication is clear. The window for building governance architecture that shapes AI development practice (rather than merely auditing its outcomes) is narrowing as deployment scales and development pipelines calcify around existing practices. The data layer will not govern itself. The frameworks that will define AI sustainability accountability over the next decade are being written now, and the decisions being made in those drafting processes will determine whether future governance is lifecycle-complete or lifecycle-partial, data-integrated or data-blind, jurisdictionally coordinated or jurisdictionally fragmented. The argument here is that only the first option in each of those pairs is equal to the challenge.

References

The following references are to be formatted in full citation style prior to publication. Key sources drawn upon in this paper include:

- Patterson, D., et al. (2021). Carbon and the Cloud. *Communications of the ACM.*
- Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and Policy Considerations for Deep Learning in NLP. *ACL*.
- Luccioni, A., et al. (2023). Estimating the Carbon Footprint of BLOOM. *Journal of Machine Learning Research*.
- Jegham, N., et al. (2025). How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference. *arXiv:2505.09598*.
- Ligozat, A., et al. (2022). Unraveling the Hidden Environmental Impacts of AI Solutions for Environment. *Patterns*.

- Kaack, L., et al. (2022). Aligning Artificial Intelligence with Climate Change Mitigation. *Nature Climate Change*.
- Verdecchia, R., et al. (2023). A Systematic Review of Green AI. *WIREs Data Mining and Knowledge Discovery*.
- Jakubik, J., et al. (2024). Data-Centric Artificial Intelligence. *Business & Information Systems Engineering*.
- Pushkarna, M., Zaldivar, A., & Kjartansson, O. (2022). Data Cards: Purposeful and Transparent Dataset Documentation for Responsible AI. *ACM FAccT*.
- Hacker, P., et al. (2024). AI, Climate, and Regulation: From Data Centers to the AI Act. *arXiv:2410.06681*.
- Bommasani, R., et al. (2024). Foundation Model Transparency Index. Stanford University.
- International Energy Agency (2024). *Electricity 2024: Analysis and Forecast to 2026*.
- Greenhouse Gas Protocol (2011, updated 2024). *Corporate Value Chain (Scope 3) Accounting and Reporting Standard*.
- NVIDIA (2025). *Product Carbon Footprint Summary: HGX H100*.
- NewClimate Institute & Carbon Market Watch (2025). *Corporate Climate Responsibility Monitor*.
- Regulation (EU) 2024/1689 (EU AI Act), Official Journal of the European Union.
- California SB 53 (2025), Transparency in Frontier Artificial Intelligence Act.
- ICAO (2016, updated 2022). *CORSIA Standards and Recommended Practices*, Annex 16, Volume IV.
- Ludvigsen, K.G.A. (2023). The Carbon Footprint of GPT-4. *Towards Data Science*.
- Cornell University / Dodge, J., et al. (2024). Projected AI CO_2 Emissions to 2030.
- Carbone 4 (2024). *Cloudy with a Chance of Hidden Emissions*.
- Heinrich Böll Stiftung (2024). The EU AI Act and Environmental Protection: The Case for a Missed Opportunity.
- *Nature Climate Change* (2020). Hiding Greenhouse Gas Emissions in the Cloud.

Author's original contributions: AI Sustainability Regulatory Framework (2025); Three-Tier AI Training Data Licensing System (2025).